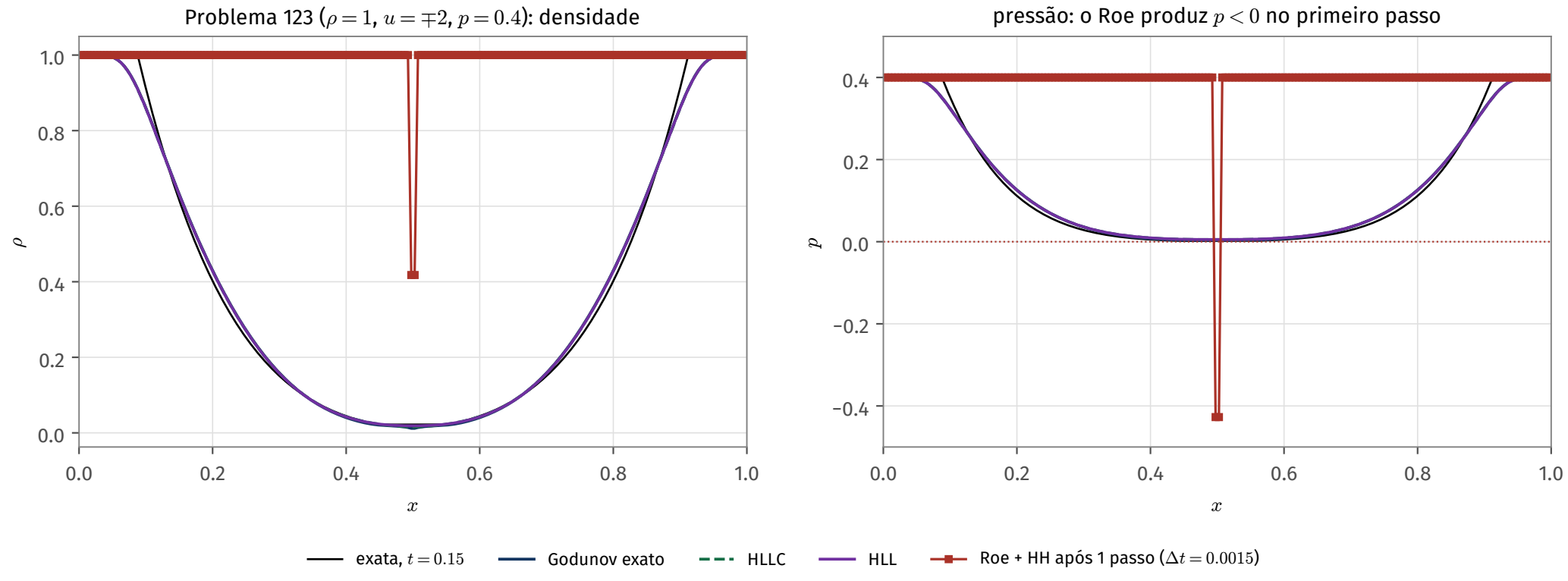
Problema 123 ($\rho = 1$, $u = \mp 2$, $p = 0.4$): densidade
pressão: o Roe produz $p < 0$ no primeiro passo
ρ
p
x
x
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
0.0
0.2
0.4
0.6
0.8
1.0
exata, $t = 0.15$
Godunov exato
HLLC
HLL
Roe + HH após 1 passo ($\Delta t = 0.0015$)